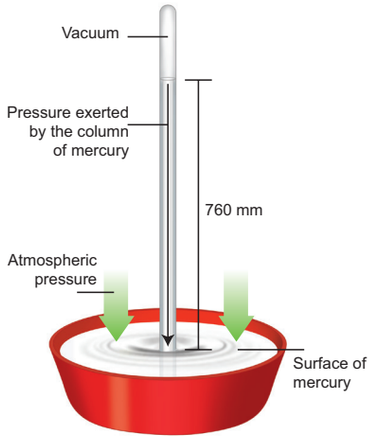


Figure 3.6 Mercury barometer

The barometer works by balancing the mercury in the glass tube against the outside air pressure. If the air pressure increases, it pushes more of the mercury up into the tub and if the air pressure decreases, more of the mercury drains from the tube. As there is no air trapped in the space between mercury and the closed end, there is vacuum in that space. Vacuum cannot exert any pressure. So the level of mercury in the tube provides a precise measure of air pressure which is called atmospheric pressure. This type of instrument can be used in a lab or weather station.

On a typical day at sea level, the height of the mercury column is 760 mm. Let us calculate the pressure due to the mercury column of 760 mm which is equal to the atmospheric pressure. The density of mercury is 13600 kg m^{-3}.

Pressure, $P = h\rho g$

$= (760 \times 10^{-3} m) \times (13600\ kgm^{-3}) \times (9.8\ ms^{-2})$

$= 1.013 \times 10^5$ Pa.

This pressure is called one atmospheric pressure (atm). There is also another unit called (bar) that is also used to express such high values of pressure.

$1\ atm = 1.013 \times 10^5$ Pa.

$1\ bar = 1 \times 10^5$ Pa.

Hence, 1 atm = 1.013 bar.

Expressing the value in kilopascal gives 101.3 k Pa. This means that, on each 1 m^2 of surface, the force acting is 1.013 k N.

Problem 3

A mercury barometer in a physics laboratory shows a 732 mm vertical column of mercury. Calculate the atmospheric pressure in pascal. [Given density of mercury, $\rho = 1.36 \times 10^4\ kg\,m^{-3}$, $g = 9.8\ m\ s^{-2}$]

Solution:

Atmospheric pressure in the laboratory,

$P = h\rho g = 732 \times 10^{-3} \times 1.36 \times 10^4 \times 9.8$

$= 9.76 \times 10^4$ Pa (or) 0.976×10^5 Pa

3.3.2 Gauge pressure and absolute pressure

Our daily activities are happening in the atmospheric pressure. We are so used to it that we do not even realise. When tyre pressure and blood pressure are measured using instruments (gauges) they show the pressure over the atmospheric pressure. Hence, absolute pressure is zero-referenced against a perfect vacuum and gauge pressure is zero-referenced against atmospheric pressure.

For pressures higher than atmospheric pressure,

absolute pressure = atmospheric pressure + gauge pressure

For pressures lower than atmospheric pressure,

absolute pressure = atmospheric pressure – gauge pressure

We have seen that liquid column exerts pressure. So the pressure inside the sea will be more. This is more than twice the atmospheric pressure. Parts of our body, especially blood vessels and soft tissues cannot withstand such high pressure. Hence, scuba divers always wear special suits and equipment to protect them (Fig. 3.7).



Figure 3.7 Scuba divers with special protecting equipment

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